

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026 — ANSWER KEY (Version 1)

Biology — Class IX | Paper: SSC-I

Syllabus: Ch 1 — The Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle | Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec 7.1-7.6)

SECTION A — Multiple Choice Questions: Answers & Explanations

Q1. Most important step of the biological method: C — Experimentation 1 mark

The textbook states plainly that **experimentation is the most important step** of the biological method, because experiments are performed to prove whether the hypothesis is true or not. The deductions drawn from a hypothesis are put to rigorous testing, and it is through experimentation that a biologist learns which hypothesis is correct. Observation, formulation of a hypothesis and summarization of results are all necessary steps, but none of them decides whether the hypothesis is right.

Q2. Characteristic that an unfalsifiable statement fails: D — It should be testable and potentially falsifiable 1 mark

A hypothesis must be **testable and potentially falsifiable**, that is, it must be possible in principle to design an experiment whose result could show the hypothesis to be wrong. A statement that no experiment could ever contradict is outside science, no matter how general or tentative it is. The other three characteristics (general statement, tentative idea, agreement with available observations) are satisfied by such a statement.

Q3. Prions are composed of: B — Proteins only 1 mark

The textbook's box on acellular particles states that **prions are composed of proteins only**, while **viroids are composed of circular RNA only**. Both cause infectious diseases in certain plants, both are acellular particles, and neither is included in any kingdom of classification. Option A describes viroids and option D describes a virus.

Q4. Unicellular prokaryotes in the five-kingdom system: C — Monera 1 mark

The five-kingdom system includes **Monera, Protista, Fungi, Plantae and Animalia**. Kingdom **Monera** holds the unicellular prokaryotic organisms. Protista holds unicellular eukaryotes, Fungi holds eukaryotes with a cell wall that cannot photosynthesize, and Plantae holds multicellular photosynthetic organisms.

Q5. NOT present in a prokaryotic cell: D — Membrane-bound nucleus 1 mark

A prokaryotic cell has no **membrane-bound nucleus**; its genetic material lies free in the cytoplasm. This is exactly what separates the prokaryotic domains Archaea and Bacteria from domain Eukarya. Prokaryotes do possess a cell membrane, cytoplasm and ribosomes, although prokaryotic ribosomes are smaller than eukaryotic ones.

Q6. Bivalents line up at the equatorial plate in: B — Metaphase I 1 mark

Bivalents (the paired homologous chromosomes formed by synapsis in prophase I) move to the equatorial plate of the spindle during **metaphase I**. In anaphase I the homologous chromosomes have already separated and are moving apart. In metaphase II the chromosomes line up *separately*, not as pairs, because the homologues were separated in meiosis I.

Q7. Tumour of normal cells that stays in one place: C — Benign 1 mark

A **benign** tumour consists of normal cells and remains in its place, so it does not invade other parts of the body. A **malignant** tumour is the cancerous kind, whose cells can leave the tumour and spread. "Metastatic" describes cells that have already spread, and "cancerous" is a synonym for malignant.

Q8. Two or more tissue types working together form: B — Organ 1 mark

In the levels of organization, an **organ** is defined as two or more types of tissues that work together to complete a specific task; the heart, liver and stomach are the textbook's examples. A group of organs performing related functions is an organ system, and several organ systems functioning together make an organism.

Q9. Plant tissue that covers and protects: C — Dermal tissue 1 mark

Plant tissues differentiate into three main types: **dermal, ground and vascular**. **Dermal tissue** covers and protects the plant. Ground tissue is the site of photosynthesis, and vascular tissue transports water, minerals and food. Meristematic tissue is not one of the three main types named here.

Q10. Disaccharide present in milk: C — Lactose 1 mark

Lactose is milk sugar, found naturally in milk and dairy products; it is formed by the condensation of galactose and glucose. The milk of all mammals contains 2 to 8% lactose. Sucrose is cane sugar, maltose is malt sugar, and glycogen is a polysaccharide, not a disaccharide.

Q11. Copying a DNA base sequence into mRNA: B — Transcription 1 mark

Transcription is the first step of gene expression, in which an RNA molecule is formed from DNA in the nucleus. Translation is the second step, in which the mRNA message is read at the ribosome to build a polypeptide. Replication is the copying of DNA into DNA, and condensation is the reaction that joins smaller molecules with the release of water.

Q12. A detachable inorganic ion needed by an enzyme: A — Activator 1 mark

Co-factors are of three types. An **activator** is a detachable *inorganic* ion such as Zn^{++} , Fe^{++} , Cu^{++} or a chloride ion; salivary amylase activity is increased by chloride ions. A coenzyme is a detachable *organic* molecule (NAD, coenzyme A, vitamins A and C), a prosthetic group is an organic molecule tightly or permanently attached (FAD, the haem group), and an inhibitor slows or stops a reaction rather than enabling it.

SECTION B — Short Questions (11 × 3 = 33 Marks)

2(i) State the FOUR characteristics that a hypothesis should have. 3 marks

1. It should be a **general statement**.
2. It should be a **tentative idea**.
3. It should **agree with the available observations**.
4. It should be **testable and potentially falsifiable**.

All four correct = 3 marks | any three = 2 marks | any two = 1 mark

OR

Qualitative vs quantitative observations, with examples, and which is more accurate. 3 marks

Qualitative observations are based on some quality or characteristic, not on a measurement. Example: the leaves of this plant are pale yellow-green, or the surface of the leaf feels rough. 1 mark

Quantitative observations are based on a measurable value and can be expressed in terms of numbers. Example: this plant is 32 cm tall, or this leaf is 6 cm long. 1 mark

Quantitative observations are more accurate, because a measured value is **invariable** and can be expressed as a number, so it means the same thing to everybody, whereas a quality such as "pale" or "rough" depends on the observer's judgement. 1 mark

2(ii) TWO living and TWO non-living characteristics of viruses, and why viruses are in no domain or kingdom. 3 marks

Living characteristics (any TWO): they occur in different varieties; they have their own genetic material in the form of either RNA or DNA; they reproduce using the material of the host cell they infect; they enter the cells of living organisms and cause diseases. 1 mark

Non-living characteristics (any TWO): they lack a cellular structure and an enzyme system; they can be crystallized and stored in a bottle; they do not respire; outside the host they behave as non-living, inert infectious particles. 1 mark

Because viruses show characteristics of **both** living and non-living things, they sit at the **borderline of living and non-living**. They therefore cannot be fitted into either side of the modern scheme, and so are **not included in any domain or kingdom** of classification. 1 mark

OR

What are prions and viroids, what is each composed of, and one characteristic shared with viruses. 3 marks

Prions are acellular infectious particles composed of **proteins only**. 1 mark

Viroids are acellular infectious particles composed of **circular RNA only**. Both prions and viroids cause infectious diseases in certain plants. 1 mark

Shared characteristic: their **infectious nature**. Like viruses they are acellular particles, and like viruses they are not included in any kingdom of the classification system. 1 mark

2(iii) Name the FIVE kingdoms; which contains prokaryotes; which contains eukaryotes with a cell wall that cannot photosynthesize. 3 marks

The five kingdoms are **Monera, Protista, Fungi, Plantae and Animalia.** 1 mark

Prokaryotes are placed in kingdom **Monera.** 1 mark

Eukaryotes that have a cell wall but cannot perform photosynthesis are placed in kingdom **Fungi.** They are mostly saprotrophic and absorb their food. 1 mark

OR

Define saprotroph, autotroph and heterotroph, and name a kingdom of Eukarya showing each. 3 marks

Saprotrophs feed on dead and decaying organic matter, which they digest outside the body and then absorb.

Kingdom: **Fungi.** 1 mark

Autotrophs make their own organic food from simple inorganic materials, for example by photosynthesis.

Kingdom: **Plantae.** 1 mark

Heterotrophs cannot make their own food and take in ready-made organic food from other organisms, usually by ingestion. Kingdom: **Animalia.** 1 mark

Accept Protista for any one of the three, since Protista shows various modes of nutrition.

2(iv) The THREE types of muscle cell, where each is found, and the protein that fills them. 3 marks

The three types shown in Fig 3.21 are **smooth muscle cells, striated muscle cells** and **cardiac muscle cells.** 1 mark

Where each is found: 1 mark

- **Smooth muscle:** in the throat and in the wall of the gut, where it produces the peristaltic contractions that move food along.
- **Striated muscle:** in the muscles of the arms and legs, which move the body parts.
- **Cardiac muscle:** in the heart.

Muscle cells have an elongated shape and are filled with **actin** and its associated contractile proteins, which is what allows them to contract and relax. 1 mark

OR

TWO structural features of a red blood cell and how each helps it transport oxygen. 3 marks

Any TWO of the following, each with its explanation: 1½ marks each

- **Biconcave disc shape:** this shape provides extra surface area, so the cell can absorb and release oxygen faster and in greater amounts than a sphere of the same volume could.
- **Nucleus, mitochondria and endoplasmic reticulum are absent:** removing these organelles frees up internal space, which lets the cell accommodate more **haemoglobin**, the pigment that actually binds the oxygen. More haemoglobin means more oxygen carried per cell.
- **Very flexible:** the cell can bend and squeeze so that it passes easily through the narrow blood capillaries, and so can deliver oxygen right up to the tissues.

The textbook also notes that red blood cells are filled with haemoglobin and that their average age is 120 days.

2(v) Why prophase I is the longest phase of meiosis, and THREE events that take place in it. 3 marks

Prophase I is the **longest phase of meiosis** because a great deal has to be completed in it: the chromosomes must condense fully, the homologous chromosomes must find each other and pair along their whole length, and genetic material must actually be exchanged between them. None of the later phases has this much to do. **1 mark**

Any THREE of the following events: **2 marks**

- Chromosomes begin to appear as thicker fibres because the chromatin coils up.
- Homologous chromosomes line up point to point against each other and form pairs, their centromeres in the same position. This pairing is called **synapsis** and each pair is called a **bivalent**.
- Non-sister chromatids of the homologous chromosomes join each other at points called **chiasmata**, and segments are exchanged between them. This exchange is **crossing over**.
- By the end of prophase I the chromosomes are fully condensed and the paired chromosomes repel each other, although they remain attached at the chiasmata.
- The nucleoli and the nuclear envelope disappear, and the spindle apparatus is formed.

OR

Metaphase I and anaphase I, and the chromosomes per pole when $2n = 12$. 3 marks

Metaphase I: the bivalents move to the equatorial plate of the spindle. One member of each homologous pair is attached to a spindle fibre coming from one pole, and the other member to a fibre coming from the opposite pole.

1 mark

Anaphase I: the two homologous chromosomes of each pair separate from each other and move to opposite poles. Each chromosome still consists of two chromatids, and each pole receives a haploid set of chromosomes. **1 mark**

If the parent cell has $2n = 12$, then the haploid number is $n = 6$, so **each pole receives 6 chromosomes** at the end of anaphase I. **1 mark**

2(vi) Identify the stages A to E on the cell cycle figure (Fig 4.3). 3 marks

Label	Stage	How it is recognized in the figure
A	Metaphase	Chromosomes aligned in a single row at the equator of the spindle.
B	Anaphase	Chromatids separated and being pulled apart towards opposite poles.
C	Telophase	Two groups of chromatids have reached the poles and are uncoiling, with nuclear membranes re-forming.
D	Prophase	Chromatin has coiled into visible chromosomes and the early mitotic spindle is forming.
E	Interphase	A single large nucleus of duplicated chromatin, labelled 46 chromosomes; the cell has not begun dividing.

The order of the cycle is therefore **E → D → A → B → C**, ending in two daughter cells with 46 chromosomes each.

All five correct = 3 marks | four correct = 2 marks | two or three correct = 1 mark

OR

Benign vs malignant tumour, metastasis, and what goes wrong in the cell. 3 marks

A **benign tumour** is a mass of dividing cells that are **normal cells** and that **stay in one place**, so the tumour does not invade the rest of the body. **1 mark**

A **malignant tumour** is the **cancerous** kind. Its cells are abnormal and are able to leave the original tumour and travel to other parts of the body. **1 mark**

Metastasis is that spreading of cancerous cells from the original tumour to other parts of the body. The underlying fault is in the cell itself: cells whose **cell cycle control system** does not work properly divide **excessively and uncontrollably**, and this uncontrolled division is what produces a tumour in the first place.

1 mark

2(vii) The FIVE levels of biological organization with one example at each level. 3 marks

Level	What it is	Example
Cellular level	Cells are the smallest unit of life, enclosed by a membrane or cell wall.	Muscle cell, skin cell, neuron
Tissue level	Groups of cells with similar functions.	Muscle, epithelial, connective tissue
Organ level	Two or more types of tissue working together for a specific task.	Heart, liver, stomach
Organ system level	A group of organs performing related functions.	Digestive system, circulatory system
Organism level	Several organ systems functioning together.	Human

2 marks for naming all five levels in the correct order | 1 mark for correct examples

OR

The FOUR types of animal tissue and one function of each. 3 marks

- **Muscle tissue:** contracts and relaxes to produce movement of the body and of materials inside it.
- **Epithelial tissue:** covers and lines the body surfaces and the surfaces of organs, giving protection and forming a barrier.
- **Nerve tissue:** conducts nerve impulses, and so carries out coordination and control in the body.
- **Connective tissue:** joins, supports and binds the other tissues and organs together.

All four named with a correct function = 3 marks | ¾ mark per correct pair

2(viii) The three main plant tissue types and the function of each. 3 marks

Dermal tissue: covers and protects the plant, forming its outer covering. 1 mark

Ground tissue: serves as the site of photosynthesis, and also stores food and provides support. 1 mark

Vascular tissue: transports materials through the plant. Xylem carries water and dissolved minerals, and phloem carries prepared food. 1 mark

OR

Why the stomach is an organ, and the tissue types in its layered walls (Fig 5.2). 3 marks

A **tissue** is only a group of similar cells performing a similar function. The stomach is an **organ** because it is made of **two or more different types of tissue organized together** to serve one particular function, namely holding and digesting food. 1½ marks

Fig 5.2 labels the layers of the stomach wall as **epithelial tissue**, **connective tissue** and **muscle tissue** (two muscle layers are shown). The textbook adds that **most organs contain all four tissue types**, the fourth being **nerve tissue**. 1½ marks

2(ix) The monosaccharide units of sucrose, maltose and lactose, and what else is released. 3 marks

(a) **Sucrose** = glucose + fructose. $C_6H_{12}O_6 + C_6H_{12}O_6 \rightarrow C_{12}H_{22}O_{11} + H_2O$ 1 mark

(b) **Maltose** = glucose + glucose (two glucose molecules). 1 mark

(c) **Lactose** = galactose + glucose. 1 mark

In each case the two monosaccharides join by **condensation**, and one molecule of **water** is released each time the bond forms. Required for full credit; may be stated once for all three

OR

Cellulose: one function in a plant, one use to humans, and why it matters in the diet. 3 marks

Function in a plant: cellulose is the structural material of the **plant cell wall**. 1 mark

Use to humans: cellulose is obtained from plants and cotton is pure cellulose, so it is used for textiles. It is also used in the **paper industry**. 1 mark

Why it still matters in the diet: the cellulose in the cell walls of fruit, vegetables, bran and whole grains is **not digestible** by humans, so it acts as **fibre or roughage**. It helps food move efficiently through the digestive tract and is effective in avoiding constipation. Cellulose fibre also absorbs toxic chemicals that may be present in food, helps to lower blood cholesterol and helps regulate blood pressure. 1 mark

2(x) Describe transcription: where it occurs, the template, and the product. 3 marks

Transcription is the first step of gene expression, in which an RNA molecule is formed from DNA. It takes place in the **nucleus** of the cell. 1 mark

Both DNA strands of the gene **unwind** from each other, but only **ONE strand of the DNA, the coding strand**, acts as the template and is copied. 1 mark

The nucleotide sequence of that coding strand is copied in the form of **messenger RNA (mRNA)**, which is the product. After the gene has been copied, both DNA strands coil together again into their normal double-stranded shape. 1 mark

OR

Describe translation: where it occurs, the role of tRNA, and the product. 3 marks

Translation takes place in the **cytoplasm**. The mRNA synthesized in the nucleus moves out into the cytoplasm, where it binds to a **ribosome**. 1 mark

Role of **tRNA (transfer RNA)**: it **transfers amino acids to the ribosome** according to the sequence of codes on the mRNA. Different types of tRNA carry different amino acids, so the right amino acid arrives for each code.

1 mark

The ribosome reads (translates) the codes carried by the mRNA and links the amino acids together, producing a **polypeptide chain with a specific sequence of amino acids**, that is, a protein. 1 mark

2(xi) The THREE types of co-factor with one example of each. 3 marks

A co-factor is the non-protein component that many enzymes need in order to function properly. There are three types: 1 mark each

1. **Activator**: a *detachable inorganic ion*. Examples: Zn^{++} , Fe^{++} , Cu^{++} , chloride ions. Salivary amylase activity is increased by chloride ions.
2. **Prosthetic group**: an *organic molecule tightly or permanently attached* to the enzyme. Examples: FAD, the haem group.
3. **Coenzyme**: a *detachable organic molecule*. Examples: NAD, coenzyme A, vitamin A, vitamin C.

OR

Why only a very small quantity of enzyme is needed. 3 marks

An enzyme is a **catalyst**: it speeds up the reaction but is **not used up or consumed** in it, so it is not needed in the same proportion as the substrate. 1 mark

During the reaction the enzyme changes **slightly in shape** as the substrate binds at its active site to form the enzyme-substrate (ES) complex, and then the enzyme-product (EP) complex. 1 mark

Afterwards, once the products are released, the enzyme **regains its original shape** and is unchanged, so it becomes free to bind another substrate molecule. Because it can be used over and over again, a very small amount of enzyme is able to catalyze a huge amount of substrate. 1 mark

SECTION C — Long Questions (4 × 5 = 20 Marks)

3(i) Apply the SIX steps of the biological method to the garden plants problem. 5 marks

Marking guidance: 1 mark for naming all six steps in the correct order, then 1 mark each for a workable hypothesis, a correctly formed deduction, an experiment that includes a control, and a conclusion that says what would happen if the hypothesis failed. Any sensible cause (light, water, soil minerals, salinity) is acceptable as long as the six steps hang together.

1. **Recognition of the biological problem.** The problem is stated as a question about living organisms: "Why are the plants on one side of the garden much smaller and paler than plants of the same kind on the other side?"
2. **Observation and identification.** Both sides are examined carefully using the senses and measurement.
Qualitative: the plants on one side are pale yellow-green, those on the other are dark green. *Quantitative:* the smaller plants average 12 cm and the larger ones 34 cm. It is also noticed that a wall shades the smaller plants for most of the day, while the soil and the watering are the same on both sides.
3. **Building up a hypothesis.** "The plants on the shaded side are smaller and paler because they receive less sunlight." This is a general statement, it is tentative, it agrees with the observations, and it can be tested.
4. **Drawing deductions.** Using "if ... then" logic: *IF* a shortage of sunlight is the cause, *THEN* plants of the same kind grown in full sunlight will grow taller and greener than identical plants grown in shade, when everything else is kept the same.
5. **Devising experiments.** Take a large number of seedlings of the same kind, the same age and the same size. Plant them in identical pots with the same soil, and give all of them the same amount of water and fertilizer. Keep one group in shade (the experimental set) and the other group in full sunlight (the **control set**). After a fixed period, measure the height of every plant and record the colour of the leaves in both groups.
6. **Inferring the result.** The quantitative data is arranged in tables and graphs and the result is drawn. If the plants in sunlight are consistently taller and greener, the hypothesis is supported. If there is no difference between the two groups, the hypothesis is rejected, and a fresh hypothesis (for example a shortage of minerals in that part of the soil) must be formed and tested in the same way.

OR

(a) Four characteristics of a hypothesis. (b) Deduction and "if ... then" logic. (c) Why experimentation is the most important step. 5 marks

(a) A hypothesis should be a **general statement**; it should be a **tentative idea**; it should **agree with the available observations**; and it should be **testable and potentially falsifiable**. **2 marks**

(b) A **deduction** is a logical consequence of a hypothesis. The hypothesis is taken as true and the observer works out what must follow from it, using "**if ... then**" logic. Worked example: hypothesis, "malaria is caused by a germ carried in the blood". Deduction: *IF* malaria is caused by a germ in the blood, *THEN* the blood of a malaria patient examined under a microscope should contain that germ, while the blood of a healthy person should not. That prediction is something an experiment can check. **2 marks**

(c) **Experimentation** is called the most important step because it is the only step that actually settles whether the hypothesis is true or false. The deductions drawn from the hypothesis are subjected to **rigorous testing**, and it is through experimentation that the biologist learns which hypothesis is correct. Without it a hypothesis stays no more than a plausible guess. **1 mark**

3(ii) The FOUR stages of meiosis II, the daughter cells produced, and interphase II. 5 marks

Meiosis II is similar to mitosis and has four stages: **3 marks, ¾ mark per stage**

- **Prophase II:** the chromatin coils up so that chromosomes become visible again, the spindle apparatus is organized, and the nuclear envelope disintegrates.
- **Metaphase II:** the chromosomes line up **separately** (not in homologous pairs) at the equator of the spindle, and each chromosome is attached by a spindle fibre from both poles.
- **Anaphase II:** the centromeres divide and the sister chromatids are pulled apart to opposite poles.
- **Telophase II:** the chromatids reach the poles and uncoil into chromatin fibres, and a nuclear envelope forms around each set. Cytokinesis II then divides the cytoplasm.

At the end of meiosis II there are **FOUR daughter cells**, and they are **haploid (n)**, each carrying half the chromosome number of the original parent cell. Because of the crossing over that took place in prophase I, they are not genetically identical to one another. **1 mark**

During **interphase II, NO DNA replication takes place**. The genome was already replicated in interphase I, which is why the chromosomes entering meiosis II still have two chromatids each. **1 mark**

OR

Labelled diagram of crossing over in prophase I, the chiasma, and why crossing over matters. 5 marks

Diagram (as in Fig 4.9): draw two homologous chromosomes, each already consisting of two chromatids, lying alongside each other and crossing over at one or more points. Only the **THREE** labels the question asks for are required: **homologous chromosomes**, **non-sister chromatids**, and a **chiasma** (the point where the non-sister chromatids cross and join). Marking the centromeres as well is a bonus, not a requirement.

3 marks: 2 for a recognizable diagram, 1 for the three labels

Note for the marker: the printed Fig 4.9 labels only "homologous chromosomes" and "non-sister chromatids"; it does not print the word chiasma on the figure itself. The chiasma has to be located from the text beneath the figure, so accept any clear mark on a crossing point.

A **chiasma** is the point at which the non-sister chromatids of a homologous pair join each other along their length. Each chiasma is a **site for the exchange of genetic material** between those non-sister chromatids, and the exchange itself is called crossing over. (The plural is chiasmata; the chromosomes stay attached at the chiasmata even at the end of prophase I, when they otherwise repel each other.) **1 mark**

Why it matters: crossing over produces **new combinations of genes** in the gametes, so the daughter cells are genetically different from each other and from the parent cell. Together with the independent assortment of chromosomes, this creates **genetic variation** among the members of a species, and variation is what allows a species to survive in a changing environment. **1 mark**

Note for the marker: sister chromatids are the two exactly similar copies attached at one centromere; the chromatids of two *different* homologous chromosomes are the non-sister chromatids, and it is these that cross over.

3(iii) The flow of genetic information from DNA to protein. 5 marks

DNA (in its genes) contains the information needed for the synthesis of proteins. The flow of this information, from DNA to RNA and then from RNA to protein, is known as the **central dogma of molecular biology** (the central dogma of life). Protein synthesis takes place in **two sequential steps, transcription and translation**, and together these two are called **gene expression**. 1 mark

Transcription is the first step. Both DNA strands of a gene unwind from each other, and the nucleotide sequence of **only one strand, the coding strand**, is copied in the form of an mRNA molecule. Once the gene has been copied, the two DNA strands coil together again into their normal double-stranded shape. 1 mark

Translation is the second step. The mRNA moves into the cytoplasm and binds to a ribosome. The ribosome reads the codes carried on the mRNA and links amino acids together in the sequence those codes specify, building a polypeptide chain. 1 mark

Where each takes place: transcription occurs in the **nucleus** of the cell; translation occurs in the **cytoplasm**, at the ribosome. 1 mark

The RNA molecules involved: 1 mark

- **mRNA (messenger RNA)** carries the message of the DNA from the nucleus out to the cytoplasm for protein synthesis.
- **tRNA (transfer RNA)** transfers the amino acids to the ribosome as per the sequence of codes in the mRNA.
- **(rRNA is a component of the ribosome itself, which combines with ribosomal protein to form the ribosome, the factory of proteins.)**

OR

(a) Chromosome, DNA and gene. (b) Gametes as the vehicle of heredity. (c) Why offspring resemble but are not identical to their parents. 5 marks

(a) Heredity is the transmission of characters from one generation to the next. A **chromosome** is a thread-like structure in the nucleus, formed from chromatin, which is DNA together with its histone proteins. **DNA** is the molecule that actually carries the hereditary information. A **gene** is a segment or unit of that DNA on the chromosome which carries the information for one particular character, or for one protein. So the relationship runs: chromosome contains DNA, and DNA is divided into genes; all the genes of an organism together make up its **genome**. 2 marks

(b) Gametes act as the vehicle for transferring hereditary information to the next generation. Meiosis in the germ cells produces haploid gametes, each carrying **one member of every homologous pair**, so each gamete carries one complete set of the parent's genes. At **fertilization** the male and female gametes fuse, the diploid number is restored, and the new individual receives one set of genes from each parent. In this way the hereditary instructions of both parents are handed on. 2 marks

(c) Offspring **resemble** their parents because every gene they possess came from one parent or the other, so the basic characters run from generation to generation. They are **not identical** to their parents because each offspring gets only half of each parent's genes, and the particular half differs in every gamete: **crossing over** in prophase I and the **independent assortment** of chromosomes shuffle the genes into new combinations, and which sperm fuses with which egg is a matter of chance. The result is a genuinely new combination of genes in every offspring. 1 mark

3(iv) The TWO types of enzyme inhibition, and which suits a medicine that must not be overcome by substrate. 5 marks

An **inhibitor** is a chemical substance that reacts with an enzyme but is not converted into products, and which therefore slows or stops the enzymatic reaction. Poisons, cyanide, antibiotics, drugs and accumulated products can all act as inhibitors. There are two types. **1 mark**

	Competitive inhibition	Non-competitive inhibition
Nature of the inhibitor	Similar in structure to the substrate.	Does not resemble the substrate at all.
Where it binds	At the active site itself, which it occupies so the substrate cannot get in.	At a different site called the allosteric site , away from the active site.
Effect on the active site	The active site is blocked but is not damaged.	The globular shape of the enzyme is altered, which damages the active site so the substrate can no longer bind.
Overcome by more substrate?	Yes. The effect is temporary and reversible, and can be overcome by increasing the substrate concentration, because the substrate then out-competes the inhibitor for the active site.	No. Adding more substrate does not help, because the active site itself has been distorted. The reaction stops.

1 mark for competitive | 1 mark for non-competitive | 1 mark for the reversibility contrast

Which type suits the medicine, and why: non-competitive inhibition. The requirement is that the bacterium must not be able to defeat the medicine by piling up its own substrate. A non-competitive inhibitor binds at the allosteric site and changes the enzyme's shape, so the active site stays damaged no matter how much substrate is present, and the enzyme remains switched off. A competitive inhibitor would simply be displaced from the active site as the substrate concentration rose, and the reaction would restart. **1 mark**

OR

(a) Enzyme specificity. (b) Optimum pH of pepsin and trypsin. (c) High vs low temperature. 5 marks

(a) Specificity. An enzyme is **specific** because it catalyzes only one type of reaction and acts only on one particular substrate. The reason is the **active site**: it is a cavity of a specific shape and charge, and only a substrate whose shape and charge fit it can bind. Example: **amylase** acts on **starch only**. It will not act on proteins or fats, however much of them are present, because they do not fit its active site. **2 marks**

(b) Optimum pH. **Pepsin** has an optimum pH of **2.00** and **trypsin** an optimum pH of **8.00**. The values are so different because each enzyme works in a different part of the digestive system. Pepsin works in the **stomach**, whose contents are made strongly **acidic** by gastric juice, while trypsin works in the **small intestine**, whose contents are **alkaline**. Each enzyme's globular structure and active site hold their correct shape only at the pH of the place where that enzyme has to work, so a slight change either side of the optimum reduces the rate and an extreme change denatures it. **2 marks**

(c) Temperature. Far **above** the optimum, the atoms of the enzyme **vibrate so vigorously** that the bonds holding its globular structure together break. The globular structure, and with it the active site, is damaged. This is **denaturation**, and it is **permanent**: the enzyme cannot recover even if the temperature is brought back down, because its shape has been destroyed. Far **below** the optimum, the enzyme is merely **inactive**, because there is not enough **energy of activation** for the reaction to proceed. Its structure has not been damaged, so it **regains its activity** as soon as the normal temperature is restored. **1 mark**

— End of Answer Key —

Total: Section A 12 + Section B 33 + Section C 20 = 65 Marks